



RATIONALISED QUALITY STANDARD, COMPONENT-SPECIFIC (RQSC)

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	Manufacturing Engineering (responsible MEM/ME-P or delegate)	Joe Sullivan	 9-July-2025
	Has Lifecycle Engineering identified any changes required for associated overhaul documentation? If yes, ME signatory must notify Lifecycle Engineering.		No
	Have Design and Stress/CPL approved the justification report? If yes, their signatures are not required on this RQSC		Yes
	Design		
	Stress/CPL		

INSPECTION REQUIREMENTS

Indications in any area of the component which are not mentioned in this standard, or when an accumulation of indications raises doubts as to the general standard of components they shall be referred to the Rolls-Royce Technical Authority for assessment and disposition.

Compliance with this RQSC may require an approved QCTP written in accordance with RRP 58000. The QCTP states the proposed non-destructive evaluation (NDE) methodology and its limitations. These limitations shall be considered by Materials Engineering (on behalf of Design) and Component Part Lifting (in consideration of component stress) and shall be agreed or rejected by those functions. If agreed, the QCTP becomes part of the engineering requirements for the component and shall be recorded in the component fixed process routing and on any applicable manufacturing drawings. This is generally required for Critical items, but may also be required for other items if NDE determines that the methodology will result in limitations that require approval.

This document is subject to the following Export Control Information:

Country
EU - SP

Export Classification
NOT LISTED

Date
23 June 2025

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2 PURPOSE

To define the quality standard for material condition and the associated manufacturing controls necessary to ensure that the Engineering Specification is achieved.

3 SCOPE

This standard must be applied to all materials / components where called up from the Component Definition.

4 GENERAL NOTES

- Any deviation from the material condition not specifically detailed by this standard must be referred to the ITP Aero Design Responsible Laboratories, via the non-conformance procedure, who will assess its effects on components integrity prior to judgement.
- This standard covers production components, for both casting and machining condition, and reflects process capability in as much as can be reasonable foreseen. Isolated defects in excess of this standard may be accepted on concession action; provided the defects are suitably charted and recorded.
- This standard will be reviewed following completion of the development programme with the ultimate aim to match process capability with engine running experience.
- All the dimensions that appear in this document are in millimetres unless otherwise stated.

5 CASTING AND MACHINING TECHNIQUE APPROVAL AND CONTROL REQUIREMENTS

For component approval, the relevant MSRR material specification, the requirements of this section, RRMS30003, and the condition of supply drawing shall be complied with.

The casting part process shall not be fully approved until the assessment of the final product has proved compliance with all the Engineering requirements. All applied processes on the casting up to the final machined part shall be included.

5.1 GENERAL REQUIREMENTS

- The surface condition shall be achieved by means which do not degrade the component material properties. The required surface finish and surface condition shall be stated on the Engineering drawing.
- An etched surface will not be subjected to heat treatment, nor is an etched surface suitable for engine running. Grain boundary attack or surface pitting are not acceptable.
- The manufacturing process shall not introduce harmful residual stress into the component.

5.2 NDT ACCEPTANCE STANDARD (CASTING AND FINISHED M/C PARTS)

5.2.1 General NDT Requirements

- In all cases the defect sizes quoted are the maximum allowable sizes of defect. The size of each defect is defined by its major axis.
- Spacing is defined as the major axis of the largest defect. In all cases, limits of separation refer to the extent of sound metal between the edge of one defect and the edge of the next.
- Defects less than the minimum separation apart are to be regarded as a single defect
- Every casting shall be non-destructively inspected for defects using visual to RRP58007, fluorescent penetrant to RRP58003 and radiographic techniques to RRP58006, each at 100% coverage. In addition, NDT (visual and fluorescent penetrant) shall be performed after final operations (heat treatment, cleaning...).
- The heat treatment specified in the relevant material specification can be used as a preparation for FPI at the caster. Different Heat treatment can be used with the agreement of ITP Aero Materials department.

- It is accepted that defects up to 0,25mm cannot be guaranteed to be observed by Visual Inspection.
- All defects up to 0,25 mm are acceptable and are not subject to spacing requirements. They must not form a clustered or linear pattern other than those permissible in Table 1 of this RQSC.
- Inclusions which break the surface are detectable by penetrant application. Inclusions are acceptable, within their relevant zone classification.
- Defects which put the component out of tolerance are not permitted.

5.2.2 Zoning diagram

The zoning diagram that will be used for the requirements at Visual Inspection to RRP58007, FPI to RRP58003 and Radiological Inspection to RRP58006 is shown in Figure 1.

COMPONENT ZONE DEFINITION

ZONE A – Airfoil LE and TE as defined by 3mm band

Inner Rail

Bosses

ZONE B & C –Remainder of the casting

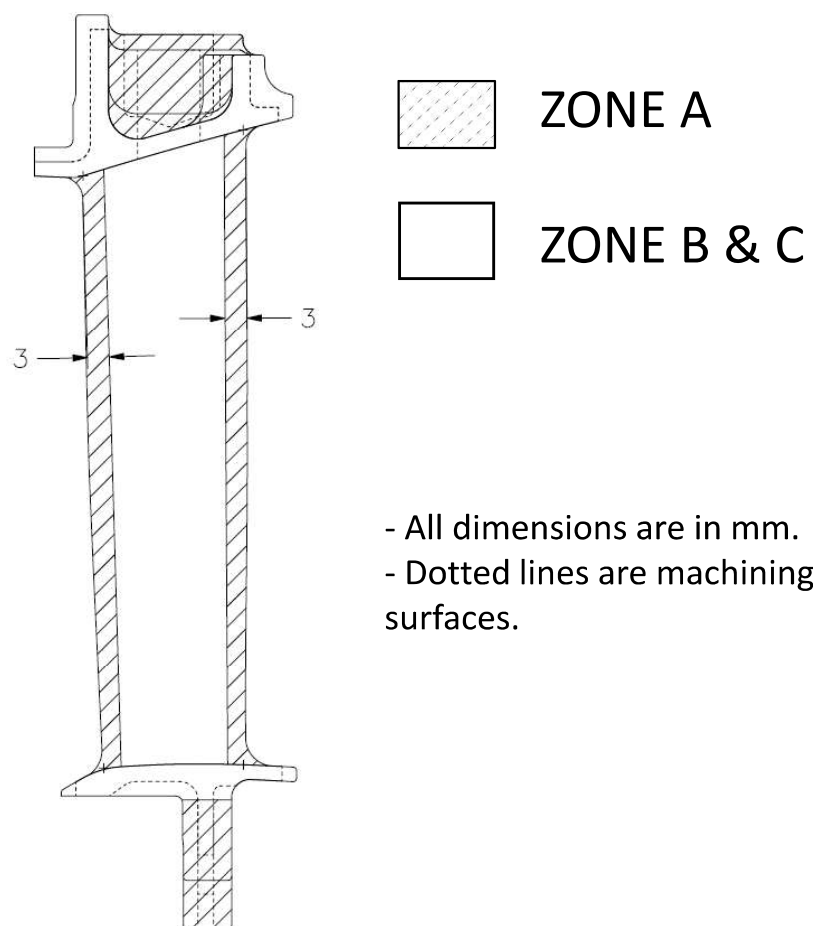


Figure 1. Defect zones (All dimensions in mm).

5.2.3 Requirements at Visual Inspection to RRP58007, FPI to RRP58003 and Radiological Inspection to RRP58006

Internal and external defect standards are defined in Table 1.

Table 1. Maximum acceptable sizes of defects.

DEFECT	ZONE A	ZONE B and C
ISOLATED	1,5mm diameter 5,0mm apart	3,0mm diameter 5,0mm apart
LINEAR	1,0mm long 5,0mm apart	2,0mm long 5,0mm apart
CLUSTERS	Maximum defect size in cluster is 1,5mm diameter Maximum number of visible defects in the cluster = 6 Minimum distance between Clusters= 5,0mm Maximum cluster size = 4,5 mm	Maximum defect size in cluster is 3,0 mm Maximum number of visible defects in the cluster = 6 Distance between clusters = 5 x Diameter of cluster Maximum cluster size = 9,0 mm
SHRINKAGE	None detected	25,0 mm diameter Distance between patches of shrinkage = 5 x diameter of shrinkage patch. Note: Any shrinkage patch shall also meet the requirements for a cluster of defects.
CRACKS	None detected	None detected

Shrinkage detected at X-ray should be assessed against the density shown in 1/8" plate 4 of ASTM E192 for dendritic in zone B and C. In all cases, this defect is defined by its diameter. If porosities bigger than 25 mm (in zone B or C) or if plate 5 or 6 (in zone B and C) repetitive defects appear during development, shrinkage acceptance criteria may be relaxed in next RQSC issue after cutup evaluation.

NOTE:

- All defects which will be removed by subsequent machining are allowable
- There shall be no defects which pass through the full section thickness
- Zones labelled "none-detected" for a particular defect shall be free of that defect above the definition of minimum size detectable at the current limits of conventional standard inspection methods

5.2.4 Visual Requirements

- A smooth transition (blend) is required in the fillet radii of all junctions.
- Reject cracks, cold shuts, linear oxide folds, nicks, dents and sharp changes of sections which due to their shape and position could propagate and cause displacement of sections of material.

5.2.5 Die Lines

Die Lines are acceptable on the gas washed surfaces (for height details see drawing notes) but never in different direction than flow one, they shall be free from laps and holes and shall not extend into the fillet radius of the aerofoil. Die lines should have a smooth transition and meet surface roughness requirement.

5.2.6 Wax Flow Lines

Wax flow lines and surface imperfections which are smooth, shallow and less than 0,25 mm deep are acceptable in all areas of the casting except the gas washed surfaces.

For gas washed surfaces, a maximum height of seal line that is mainly in longitudinal direction (from LE to TE) of 0,25 mm is acceptable due to wax sew procedure in the inner and outer platforms.

For the remaining gas washed surfaces, a maximum wax die height of 0,125 mm is accepted.

5.2.7 Acceptance Standard for Clustered Defects

The visual, penetrant and radiological requirement for clustered defects shown in Table 1 give the maximum size and number of allowable defects. The minimum spacing between any two defects in the cluster shall be equivalent to the major axis of the largest defect in the pair being considered.

5.3 MACROSTRUCTURE

- Etch operation as per RRP51002.
- Macrostructure shall be in accordance with CME 5013 Section 2. Brochure No.12A with the exceptions detailed below.
- All leading and trailing edges are to be quantitative inspected for columnar grain. Columnar grain is defined as having a minimum aspect ratio of 4:1
- Columnar grains up to 25 mm are acceptable. See Figure 2.
- Big and misoriented grains as that seen on Figure 2 are acceptable according to TRP-003745/001. Materials evaluation is provided in CSH-00018448.
- Big equiaxed grains in inner and outer platform as those seen in Figure 3 are accepted according to TRP-003745/001. Materials evaluation is provided in CSH-00018448.
- Due to differing nozzle configurations the grain size requirements detailed above may not be achievable. Therefore, equiaxed and columnar grain in excess of these limits may be acceptable subject to agreement with ITP Aero Materials Department and will be included in the RQSC.
- Photographs illustrating agreed variations to this standard based on process capability will be included in the Process Approval Report. This report will thereafter constitute the grain size acceptance standard for that particular casting technique relative to the part number/Foundry combination.

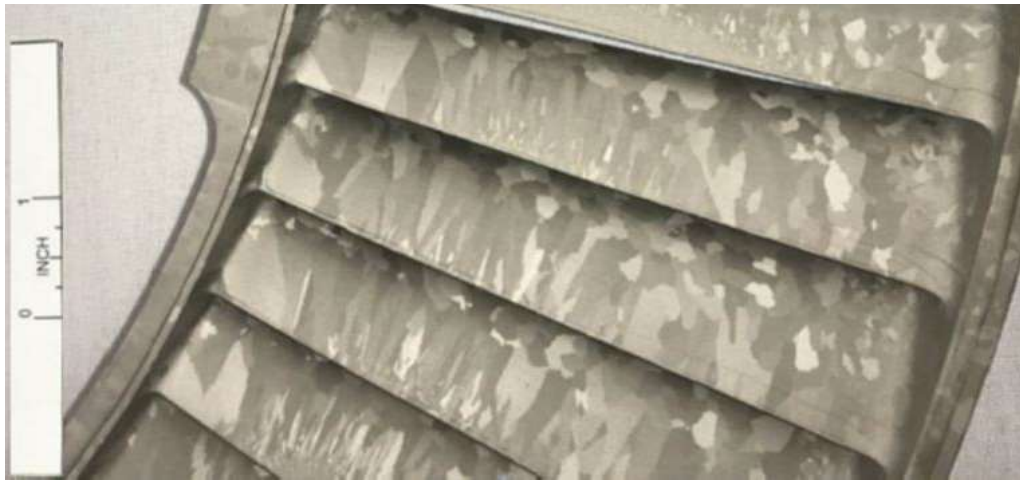


Figure 2. Big and misoriented grains.

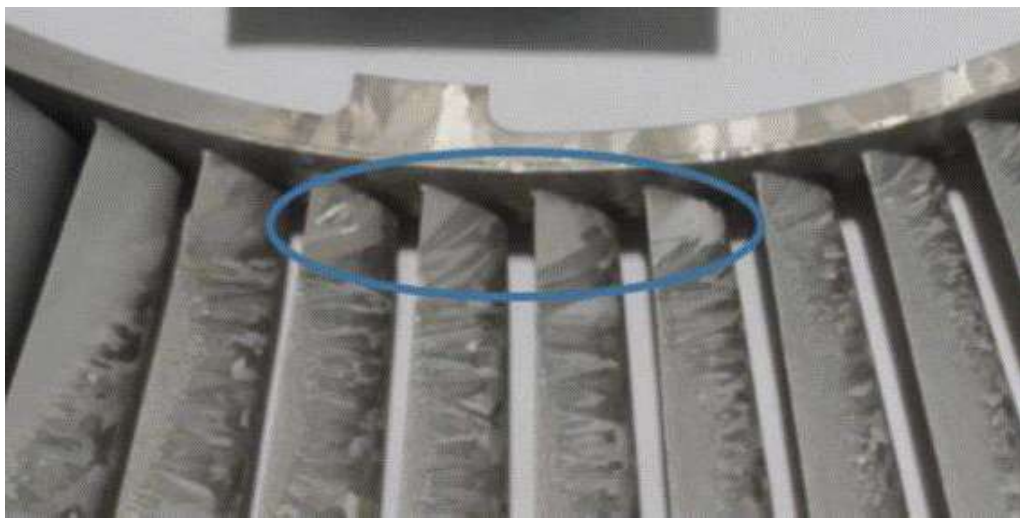


Figure 3. NGV4 big columnar and equiaxed grains.

5.4 MICROSTRUCTURE

Typical microstructure is shown in CME 5013 Section 2 Brochure 13B.

5.5 ACCEPTANCE STANDARD MICROPOROSITY

The aim of the metallographic examination is to know the microstructural quality of the part in general, at the location of the peak stresses and at the location of potential areas of defects. With this purpose, vanes have to be analysed in standard sections, at peak stress locations and in potential areas of defects.

The location of the cutups to evaluate the general condition (standard sections) and the highly stressed areas of the vane can be seen in the Figure 4.

The microporosity assessment shall be carried out using an optical microscope with a frame size of 1,25mm x 1,00mm (0.050" x 0.040"). Each micro-section shall be completely scanned to locate and record the worst area of porosity. The remaining micro-section area shall then be assessed by locating frames adjacent to the worst area of porosity until the complete micro-section has been analysed.

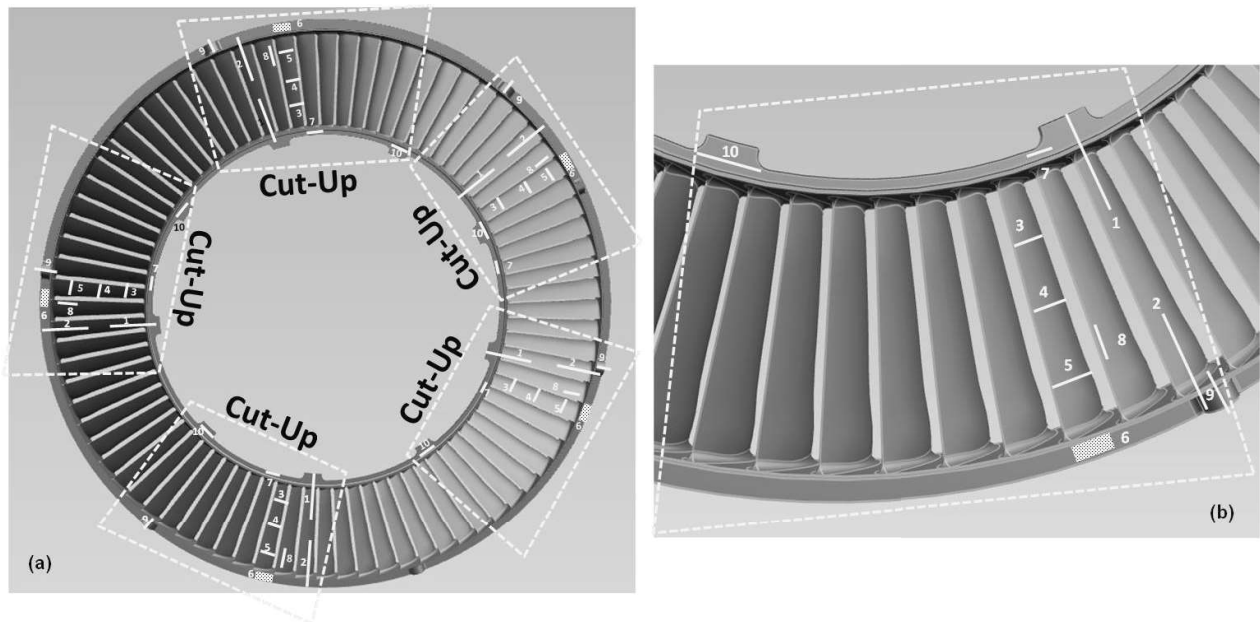


Figure 4. (a) Full Cut Up Plan Reference (b) Detailed view.

Notes:

- (a) NGV cut up plan shall be carried out as per Figure 4 (a) placing the dashed area (cut 1 to 10) described in Figure 4 (b), matching as reasonably as possible, the extremes of the part that is the closest to the largest mass of gating, or farthest away from the largest mass of gating. This information needs to be reported to Rolls-Royce Technical Authority
- (b) Each individual cut, 1 to 10, may include two or more different areas as per component definition section 5. Each to be assess against specific requirements.
- (c) Cut number 6 is to be done parallel to the surface.
- (d) Deviations to the cut-up plan need to be agreed with Rolls – Royce Technical Authority.

Before starting with the standard sections cutups, flat shots must be performed:

- 1) Cut inner platform, outer platform and aerofoils where standard cutups will be performed.
- 2) X ray inspection shall be performed in these separate sections as illustrated in Figure 5. Finally, ITP Aero Materials & Processes department shall review the X-ray results and decide a cutup plan based on these results.

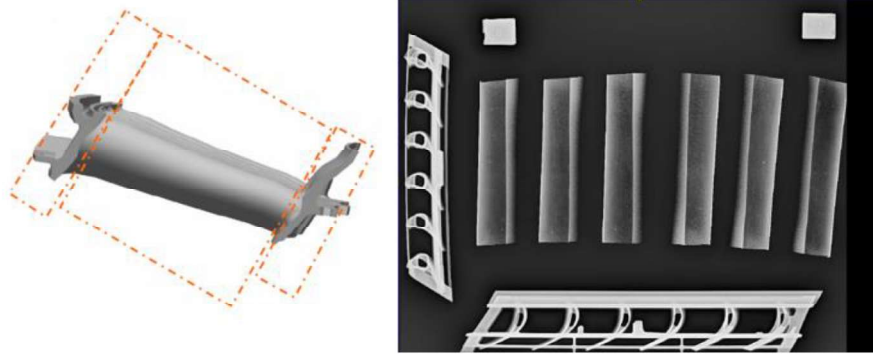


Figure 5. Sectioning plan example for close contact X-ray evaluation.

This cutup plan will be directed to those areas showing defects (mainly shrinkage) on the standard X-rays of the whole validation lot or in the X-rays of the sections of the castings submitted to close contact shots.

The maximum permissible porosity readings are presented in Table 2.

Table 2.- NGV porosity readings.

Porosity Level	NGV Zone		
	A	B	C
$3\% \leq \text{Level} \leq 4\%$	None	None	None
$2\% \leq \text{Level} \leq 3\%$	None	1	1
$1\% \leq \text{Level} \leq 2\%$	3	5	5

NOTE: Porosity levels < 1% shall be considered as non-rateable.

The above standard is provisional and may be relaxed during process validation.

6 SALVAGE / RECTIFICATION

6.1 WELDING

Salvage welding is not permitted.

6.2 DRESSING

- The surface finish of the dressed area shall be equal or better than the surfaces surrounding the dressed area and shall be free from scores. Scratches are permissible.
- Dressed areas shall have a smooth transition into sound metal.
- All defect which will be removed by subsequent machining are allowable.
- After dressing the component shall be locally re-inspected according to RRP58007 (Visual Inspection) and RRP58003 (FPI).

6.2.1 Blending allowance

- Blending the areas described in Figure 6 is permissible and shall be carried out within the requirements described in Table 3.

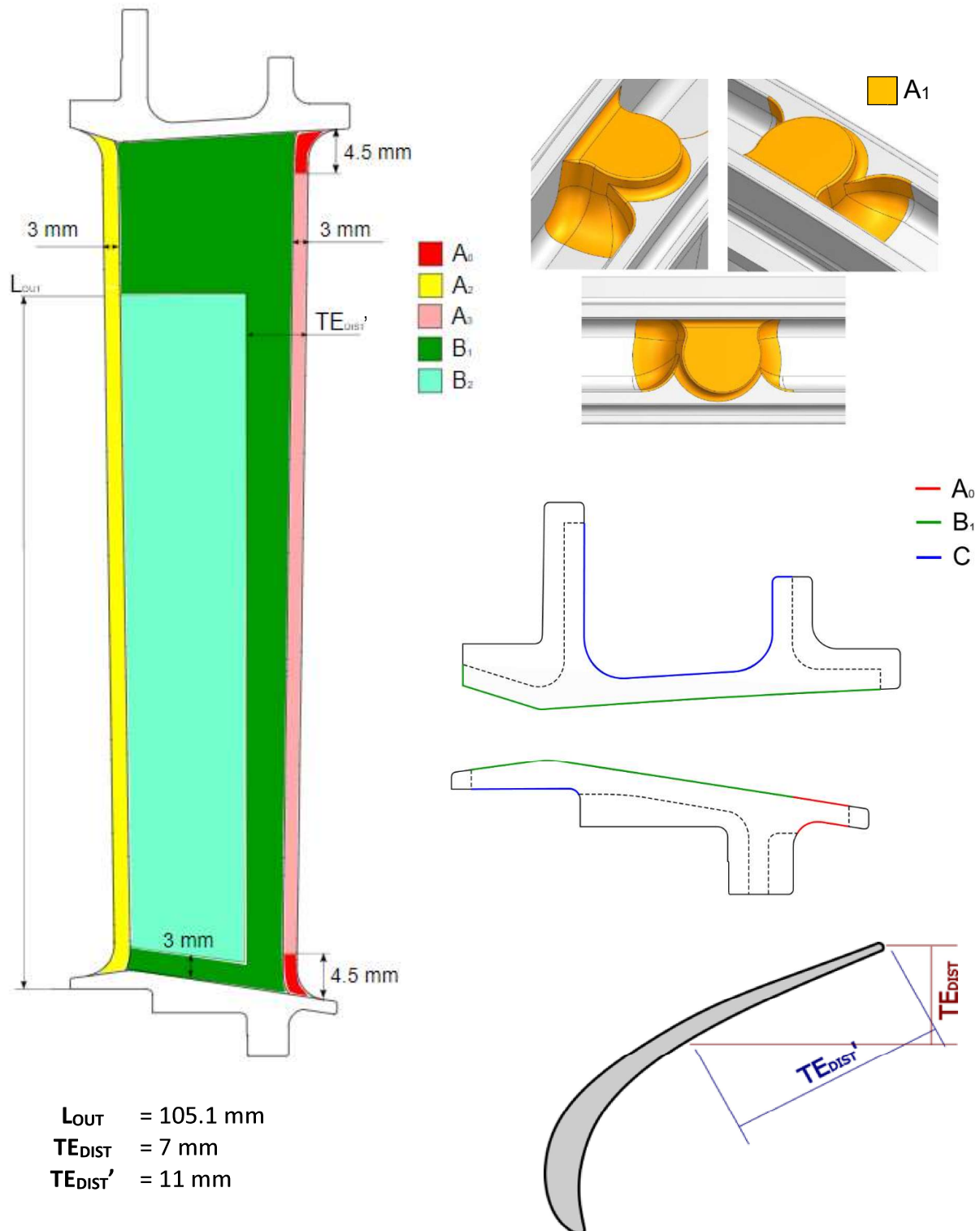


Figure 6. Defect zones (All dimensions in mm).

Table 3. Allowable blendings.

Zone	Depth	Extension	Feature	Remark	Remark 2
A0	None	None	TE & LE (3 mm) Rear inner platform (flowpath and no flowpath)	N/A	N/A
A1	0.4 mm	- Faired min 5X Depth, 6mm max - Separated 3X the max blend faired	Bosses	N/A	8 per cast
A2	0.381 mm deep or 25% of LST* whichever less	- Faired min 5X Depth, 8 mm max - Separated 3X the max blend faired	LE (3 mm) Under line defined	Maximum 6 per airfoil	Maximum 6 per airfoil (either A2 and/or A3 combination) Maximum total number of blendings 24 per castings. The combined total depth/thickness reduction of two blends aligned on opposite sides of pressure and suction surfaces may not exceed the allowance of a single blend.
A3	0.5 mm deep or up to min wall 0.535 mm	- Faired min 5X Depth, 8 mm max - Separated 3X the max blend faired	TE (3 mm) Under line defined	Maximum 6 per airfoil	
B1	0.381 mm deep or 25% of LST* whichever less	- Faired min 5X Depth, 8 mm max - Separated 3X the max blend faired	Airfoil (suction and pressure surfaces including fillets as indicated) Platforms (washed surfaces)	Maximum 6 per airfoil.	Only one blend per fillet. Maximum 6 per airfoil (all B1 or 1xB2 + 5xB1) Maximum total number of blendings 24 per castings. The combined total depth / thickness reduction of two blends aligned on opposite sides of pressure and suction surfaces may not exceed the allowance of a single blend.
B2	0.75 mm deep or 50% of LST* whichever less	- Faired min 5X Depth, 8 mm max - Separated 3X the max blend faired	Airfoil (suction and pressure surfaces indicated)	Only one blend per airfoil.	
C	0.5 mm deep	- Faired min 5X Depth, 8 mm max - Separated 3X the max blend faired	None gas washed surfaces as indicated	No blendings aligned on opposite sides of inner platform (flowpath and no flowpath) surfaces may not exceed 0.5 mm	

*LST: Lowest Section Thickness

6.2.2 Cutbacks

- Cutbacks are defined as reworks where dressing direction is perpendicular to the Zone A cross-section (3 mm band of leading and trailing edges) in a way that the airfoil chord is reduced (Figure 7).
- Cutbacks are allowed in the areas described in Figure 8 and shall be carried out within the requirements described in Table 4.
- Only one cutback per airfoil is allowed considering both leading and trailing edges.

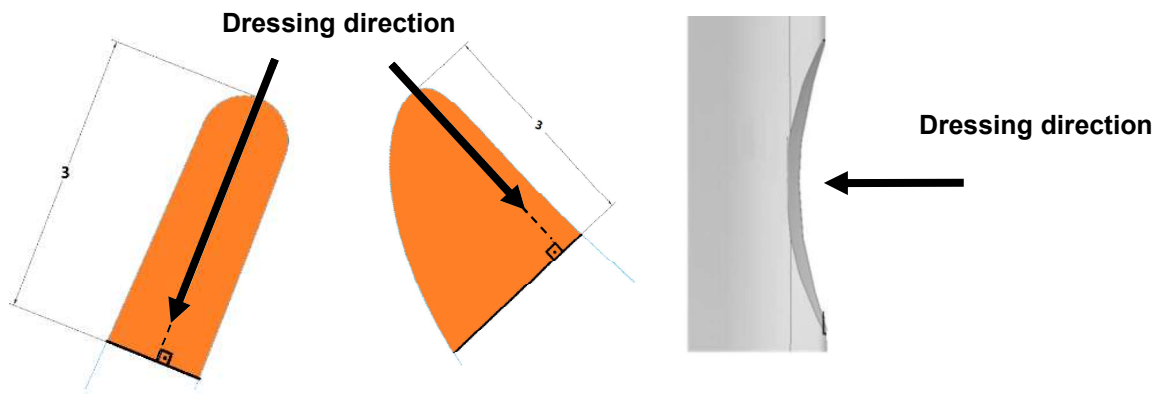


Figure 7. Cutback direction sketch.

Table 4. Allowable cutbacks.

Zone	Max Depth	Max Length	Quantity	Remark	Remark 2
Leading Edge	1 mm	10 mm	4 x NGV Ring Max 1 x Airfoil	N/A	Defects must be considered altogether. The deepest and longest defect determines the cutback allowance in terms of quantity per NGV Ring. Min acceptable length is 4X (X=depth).
	2 mm	10 mm	1 x NGV Ring		
Trailing Edge	1 mm	5 mm	12 x NGV Ring Max 1 x Airfoil	Permitted in all the airfoil span (continuous green line + green dotted line). See Fig 8. Fillets not allowed.	
		10 mm	6 x NGV Ring Max 1 x Airfoil		
		20 mm	3 x NGV Ring Max 1 x Airfoil		
		55 mm	1 x NGV Ring Max 1 x Airfoil		
	2 mm	10 mm	2 x NGV Ring Max 1 x Airfoil	Permitted up to 92 mm of the airfoil span (green dotted line). See Fig 8. Fillets not allowed.	
		20 mm	1 x NGV Ring Max 1 x Airfoil		

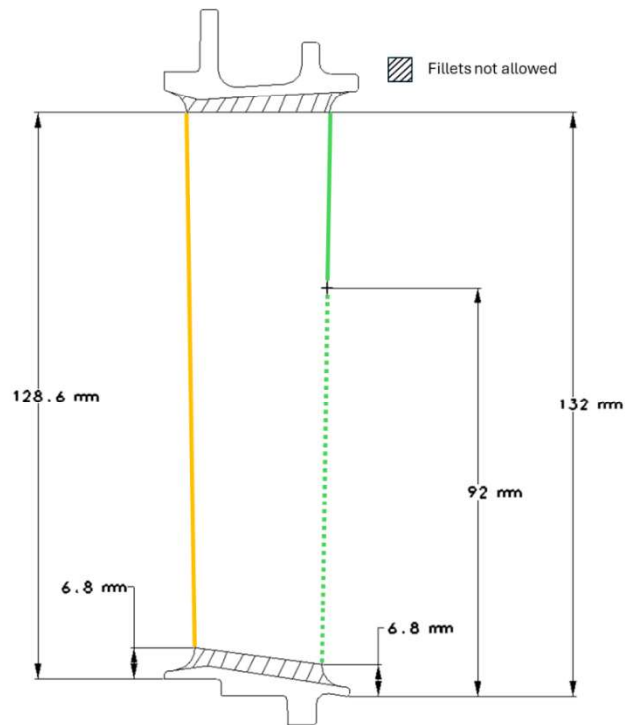


Figure 8. Cutbacks allowance areas.

6.3 DIMENSIONAL RECTIFICATION

Any dimensional corrections must be carried out in accordance with MSRR 9916.

All casting to have FPI carried out after dimensional correction.

Annulus deviations greater than 5.00 mm are not to be rectified.

7 COMPONENT APPROVAL REQUIREMENTS

Process approval shall be carried out in accordance with RRMS30003.

The validation moulds shall be subject to basic N.D.T. examination i.e. penetrant, visual inspection and 100% conventional x-ray inspection including angle shots into the shrouds and fillet radii.

Sections shall be selected in accordance with the cut-up diagram detailed in Figure 4. However, if N.D.T. examinations dictate that problem areas exist outside the cut-up plan, then additional sections shall be directed towards these regions.

8 COMPONENT RELEASE REQUIREMENTS

There are no additional release requirements beyond those specified in the material and process specifications and this quality standard.

9 DEFINITIONS

Where applicable refer to RQSG000/1.

10 REASON FOR ISSUE

This standard contains the following changes from the previous standard. The report(s) referenced confirms that the standard, or changes to the standard, have no impact on product integrity.

Page(s)	Change Summary	Report Reference
All	Justification Report	EDNS01001274063 CSH-00035664
10-11	Section 6.2.1: Zoning updated with A3 area. Table 3 and Figure 6 updated.	
12-13	Section 6.2: Table 4 updated with new allowances. Figure 8 updated.	